

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$S_{\text{eff}} = \frac{1}{4\pi\alpha'} \int d^2\xi \partial_a X_0^\mu \partial^a X_{0\mu} + \frac{g}{2\sqrt{2\alpha'}} \int_{-\infty}^{+\infty} d\tau \int d^{26}k \exp[ik \cdot X_0(\tau)] \hat{\mathcal{L}}_{\text{B}}(k, \tau) + \cdots,$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\int_{\Omega} \left(\left\langle \frac{\partial d}{\partial x_i}, \frac{\partial d}{\partial x_j} \right\rangle \frac{\partial Y^i}{\partial x_j} - \frac{|\nabla d|^2}{2} \text{div} Y + \langle \tau, Y \cdot \nabla d \rangle \right) = 0, \quad \forall Y \in C_0^\infty(\Omega, \mathbb{R}^3).$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$S_{lc}^F = \frac{i}{4\pi l_s^2} \int d\tau \int_0^{2\pi l_s^2 p^+} d\sigma \left[\left(\eta^{\alpha\beta} \delta_{AB} - \epsilon^{\alpha\beta} \sigma_{AB}^3 \right) \partial_\alpha x^+ \bar{\theta}^A \Gamma_+ (\mathcal{D}_\beta \theta)^B \right],$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\sup_{\gamma \in \mathcal{P}(S)} \left\{ \int_S g \, d\gamma - \lambda_1 \mathbf{K}_1(\mu_{13}, \gamma_{13}) - \lambda_2 \mathbf{K}_2(\mu_{23}, \gamma_{23}) : \mathbf{K}_\ell(\mu_{\ell 3}, \gamma_{\ell 3}) < \infty \ell = 1, 2 \right\},$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\Gamma_{ij}^{\mathfrak{t}} = a \left(\frac{da}{dt} \right) \gamma_{ij}, \quad \Gamma_{\mathfrak{t}j}^i = \frac{1}{a} \left(\frac{da}{dt} \right) \delta_j^i, \quad \Gamma_{jk}^i = \frac{\gamma^{il}}{2} (\partial_j \gamma_{lk} + \partial_k \gamma_{lj} - \partial_l \gamma_{jk}).$$